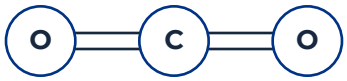
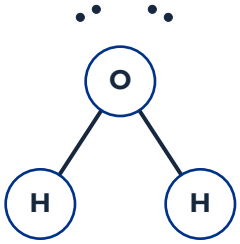
Molecular shape and polarity

Electron domains determine an arrangement; atom positions determine molecular shape.

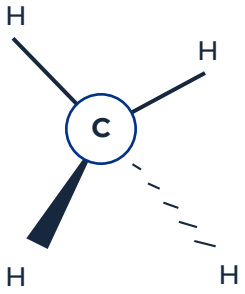
CO2



H2O



CH4



Molecule	Central domains / lone pairs	Molecular shape	Polarity
CO2	2 domains; 0 lone pairs	Linear	Nonpolar; equal opposite bond dipoles cancel.
H2O	4 domains; 2 lone pairs	Bent	Polar; bond dipoles do not cancel.
CH4	4 domains; 0 lone pairs	Tetrahedral	Nonpolar for this symmetric molecule.
NH3	4 domains; 1 lone pair	Trigonal pyramidal	Polar.

Count domains, then name shape

Each single, double or triple bond counts as one electron domain. Lone pairs count as domains but are not atoms in the molecular-shape name. Four domains give a tetrahedral electron-domain arrangement.

Ideal tetrahedral angle: about 109.5 degrees. Wedge = toward you; dashed bond = away. Molecular polarity is directional: polar bonds can cancel by symmetry. VSEPR is a model, not an electron-density map.

Molecular shape and polarity

Self-check questions and explanations. Refer to the visual guide on the front.

How many central electron domains do CO2, H2O and CH4 have?

CO2: 2 domains (two double bonds); H2O: 4 (two bonds, two lone pairs); CH4: 4 (four bonds). Count regions around the central atom, not each line in a multiple bond separately.

How do electron-domain geometry and molecular shape differ for these molecules?

CO2: linear electron geometry and molecular shape. H2O: tetrahedral electron geometry, bent molecular shape, two O lone pairs. CH4: tetrahedral electron geometry and molecular shape.

Why is H2O polar but CO2 nonpolar?

Dipoles point toward O in both examples. Equal opposite CO2 bond dipoles cancel by linear symmetry; bent H2O dipoles do not cancel, so H2O is polar.

How does NH3 compare with CH4?

NH3: 4 domains, 1 lone pair, trigonal pyramidal and polar. CH4: 4 domains, no central lone pairs, tetrahedral and nonpolar. Domains include both bonds and lone pairs.

Do polar bonds always make a polar molecule?

Molecular polarity is the vector sum of bond dipoles and the charge distribution. Direction and symmetry matter: CO2 has polar bonds but zero net dipole; H2O has a net dipole.

What do wedge/dash bonds show, and what is the ideal tetrahedral angle?

A tetrahedral sketch can use two in-plane bonds, a wedge toward the viewer and a dashed bond away. The ideal angle is about 109.5 degrees. Flat drawings can hide depth and invite incorrect planar interpretations.